

Learning comparison weights without match labels

Expectation–maximisation (EM) fits parameters using many record pairs.

Prepare training comparisons

Pair 1: (agree, agree, agree)
Pair 2: (agree, contradicts, agree)
Pair 3: (disagree, agree, disagree)
... across many training pairs

Choose starting estimates

Estimate u from random pairs.
Keep u fixed during this EM run.

Choose initial m values
and the training match fraction.

E step: score every training pair

Use the current parameters to calculate
each pair's match probability.

soft weights

Example: 80% match probability

→ 0.8 match count + 0.2 nonmatch count.

These are **fractional counts**.

M step: update the parameters

Add the fractional counts.
Normalize them to update m
and the training match fraction.

Keep separately fixed
parameters unchanged.
The estimates give new field weights.

next iteration: $t \rightarrow t + 1$

Iteration 0 uses the starting estimates. Iterations 1, 2, ... repeat E then M on the same comparisons.

Stop when parameter changes are below the tolerance or the iteration cap is reached

Retain the fitted parameters and record whether the convergence criterion was met.

After fitting: score new pairs

New record pair (a, b)

construct $\gamma(a, b)$

Apply fitted comparison weights

add prior log odds for the
stated scoring population

Match probability

then apply the declared
decision rule

Splink practice. A common route estimates u from random pairs first and holds it fixed during EM.

This notebook's EM arm uses this route. The supervised reference estimates m from known matching pairs.

A stable fit still needs independent evaluation.